Ziqi Lin

Applied AI Engineer | LLM Workflow Automation | HITL & Evaluation Systems | Python / AWS

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Work Authorization: F-1 OPT / STEM-eligible through 2029, no sponsorship required

SUMMARY

Applied AI Engineer building LLM-assisted workflow automation for compliance-sensitive operational systems. Experienced in rule-first AI workflows, structured LLM outputs, human-in-the-loop review, audit logging, and replay-based evaluation. Strong backend and data reliability foundation across Python, FastAPI, AWS, SQL, schema validation, asynchronous processing, and workflow recovery.

SKILLS

- **AI / LLM Systems**: OpenAI / Anthropic APIs, OpenRouter, structured outputs, prompt engineering, LLM-assisted extraction & normalization, multimodal AI (image / OCR / speech), LangChain, HuggingFace fine-tuning (RoBERTa)
- **LLM Tooling & Context**: tool / function calling, RAG, FAISS, context management, short-term memory, prompt / model versioning
- **Reliability & Evaluation**: human-in-the-loop review, audit logging, replay / benchmark testing, frozen-input evaluation, idempotent processing, retry / dead-letter handling, PHI / PII-aware handling
- **Backend & Cloud**: Python, FastAPI, Flask, REST APIs, async processing, JSON schema validation, Git, Docker, GitHub Actions (CI/CD), AWS (Lambda, S3, API Gateway, ECS, EventBridge, DynamoDB)
- **Data Foundations**: SQL, PostgreSQL, Redshift, dbt, Airflow, ETL/ELT, data quality, data contracts, Kafka, PySpark, Databricks

EXPERIENCE

Direct Care Home Health Services / Premier Health Group — Washington, D.C., US

Software Engineer, AI Systems & Automation — September 2025 – Present

Multimodal AI-Assisted Patient Intake & HITL Review Workflow | Public Demo | GitHub

- Built a production AI-assisted patient intake and review workflow that automates retrieval, parsing, and normalization of referral, patient, and nurse-recorded speech-to-text data with a **sub-1-minute** processing SLA, supporting ~20 referrals/week.
- Integrated conditional LLM advisory after deterministic rule evaluation to generate schema-validated summaries, risk flags, missing-field analysis, and reviewer notes, reducing reviewer validation time **from ~5 minutes to ~52 seconds** per case.
- Improved handwritten payer-field extraction quality using Enum-constrained structured outputs, reference examples, and LLM-assisted normalization for OCR-parsed referral fields, increasing measured field accuracy **from ~60% to ~95%** on recent cases.
- Implemented policy-driven **human-in-the-loop** review for uncertain, high-risk, or incomplete cases, enabling reviewer confirmation, note capture, final routing adjustments, and audit-ready decision tracking.
- Built a **multi-layer evaluation framework** spanning system-level, component-level, and human-AI interaction metrics, including historical-truth comparison, OCR field accuracy, processing latency, override rate, review time, and adoption signals.
- Engineered a replay-based evaluation workflow with a **versioned S3 case-artifact store** containing ~200 historical records across PDF, image, and audio artifacts, enabling reviewer feedback capture, historical comparison, artifact updates, and prompt/model regression testing on complete-source cases without affecting production outcomes.

Healthcare Data & Eligibility Infrastructure

- Developed a production healthcare data ingestion and eligibility verification platform on AWS with Lambda-orchestrated vendor API checks, asynchronous processing, and Redis-backed patient pre-screening, reducing verification latency **from ~3 minutes to 15–30 seconds** and eliminating next-day manual portal lookups.
- Hardened billing-critical data and downstream AI inputs with multi-layer storage, JSON schema validation, data quality checks, data contracts, idempotent processing, replay/backfill, and GitHub Actions CI/CD.

PROJECTS

Retrieval-Augmented Real Estate Email Assistant Prototype | LangChain, FAISS, RAG, GPT/Claude APIs, Flask, MySQL

- Built a retrieval-augmented LLM email assistant that combines **keyword filtering** with **FAISS vector search** to match client preferences against property, listing, and market-context materials before drafting personalized outreach.
- Implemented a generate-then-validate loop where each draft is checked via structured JSON output for preference alignment, factual consistency, and hallucination risk, with rewrite suggestions surfaced before human review.

Reddit Sentiment Pipeline with Fine-Tuned RoBERTa | HuggingFace, PyTorch, Kafka, Databricks, PySpark, Delta Lake | GitHub

- Fine-tuned RoBERTa-base for GoEmotions multi-label emotion classification, comparing full fine-tuning, partial encoder freezing, and baseline models; achieved **0.580 micro-F1, 0.489 macro-F1, and 0.921 macro ROC-AUC**, with Kafka/PySpark streaming ingestion of **100K+** Reddit events for downstream analytics.

EDUCATION

University of Maryland | College Park, MD | Master of Science in Information Systems | GPA: 3.9 | August 2024 – December 2025

Beijing Jiaotong University | Beijing, China | Bachelor of Management in Information Systems | GPA: 3.8 | August 2020 – June 2024